

2(a)	1 correct counting / correct number of divisions / 28 ± 2 ; 2 for correct conversion(s) to micrometres ; 3 correct answer / 8.96 / 8.32–9.6 ;	3
2(b)(i)	measure the, widest / narrowest / longest, part of the each nucleus ;	1
2(b)(ii)	<i>any one from:</i> 1 (same) magnification / $\times 400$; 2 (same) number of / 100 nuclei (measuring) <u>per. person.</u> ; 3 (sample from / same) thyroid / gland ; 4 same / one, scientist (measuring the diameter) ; 5 <i>idea that:</i> type of tumour cell is unknown ;	1
2(b)(iii)	there is no difference, in / between the (mean) nuclear diameters of, the Pap-stained cells and H&E- stained cells ;	1
2(b)(iv)	accept null hypothesis or there is no <u>significant</u> difference ;	1
2(b)(v)	<i>state</i> 1 <i>idea that</i> this technique can be used to distinguish between benign and malignant cells or the <u>mean nuclear diameter</u> of benign cells is smaller than in malignant cells / ora ; <i>explain</i> 2 $\text{mean} \pm 2 \text{ SE}$, do not overlap or there is a <u>significant difference in the mean nuclear diameters of benign cells and malignant cells</u> ;	2

Question	Answer	Marks
2(b)(vi)	<i>disadvantages</i> • time consuming (prepping / counting / measuring) • labour intensive • shortage of skilled scientists • invasive • only thyroid / one tissue, used • human error – due to, small values / very close values / large numbers counted <i>impact:</i> • infection / bleeding • pain • tissue damage • metastasis • anxiety • delayed treatment • unnecessary treatment • idea of misdiagnosis • possibly not feasible on large scale • may not be true of other tumours	2